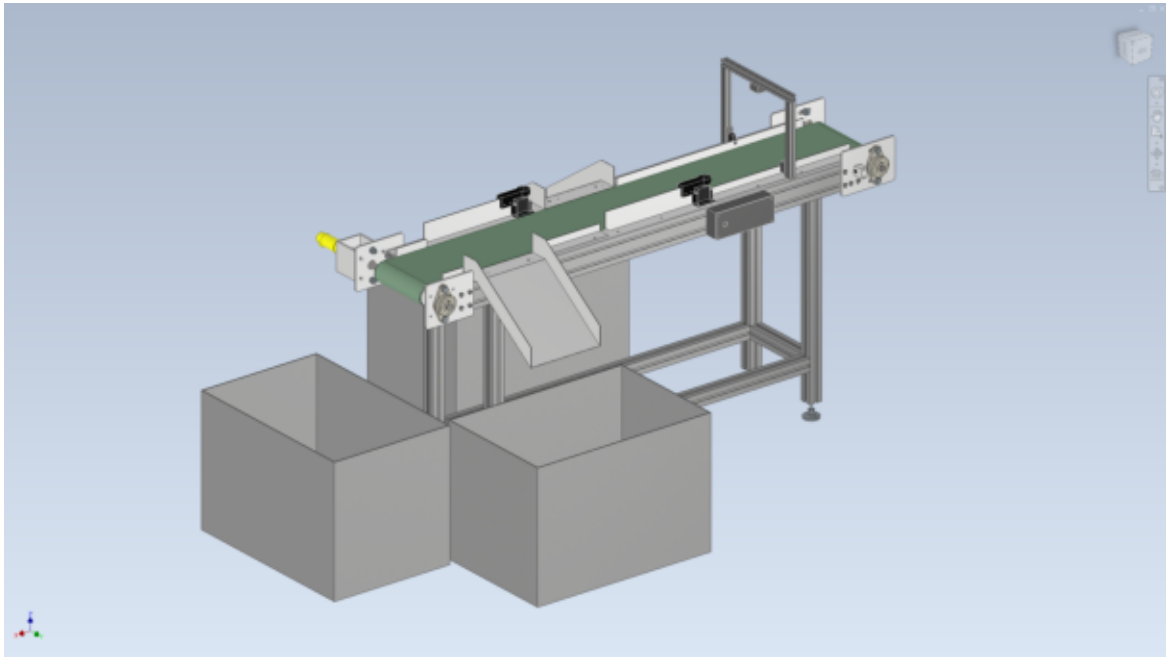
Automated Conveyor Sorting System - ConveyorSorter

ConveyorSorter - calculation report. Designed and simulated only: every value is calculated or simulated from the CAD model, nothing has been built or tested.

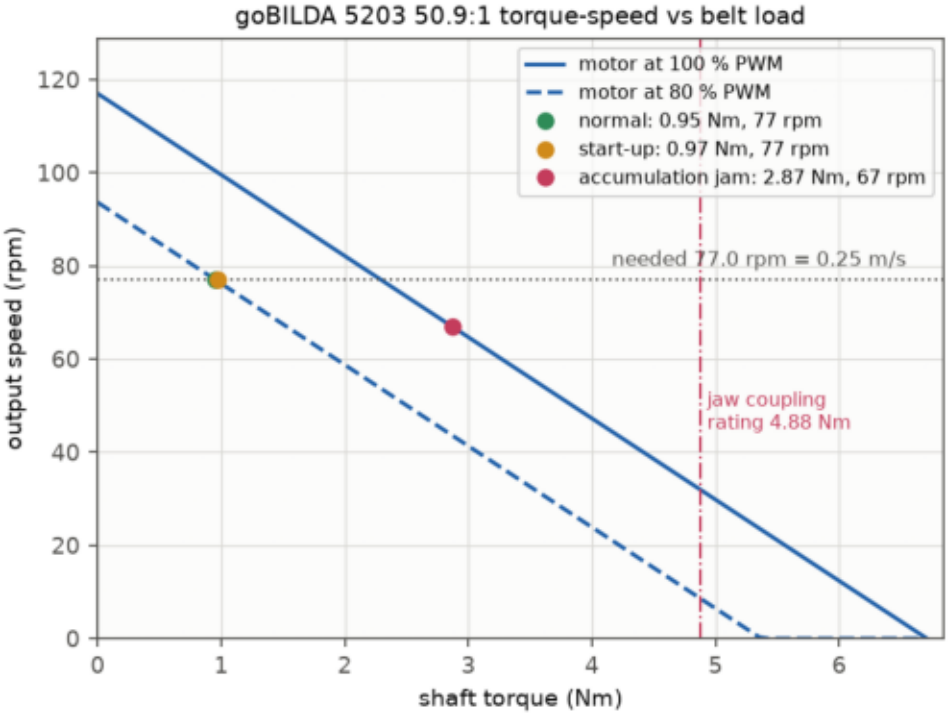


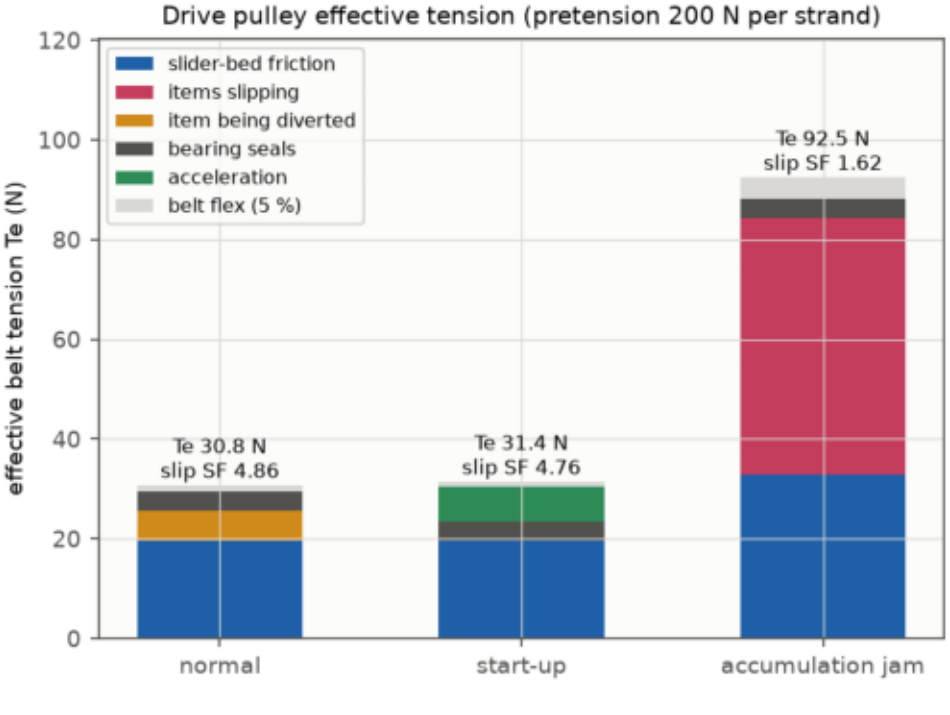
Item	Value
Belt	200 x 2 mm PVC, 3195 mm loop, 0.25 m/s, 77.0 rpm pu
Drive	goBILDA 5203 50.9:1, 0.95 Nm normal (7.0x stall margi
Items	50-180 L x <=150 W x 30-150 H mm, <= 1.5 kg; 3 clas
Sensing	through-beam + VL53L1X ToF at X 280; decision margin
Diverters	2 x Savox SB-2290SG, paddle at 40 deg, hold SF 2.15
Sort rate	1858 items/h at a 370 mm gap, 100 % mechanical succ
Assembly	104 occurrences, 309 constraints, 0 DOF, 0 clashes at 1
Mass	32.7 kg machine + 7.2 kg totes (Inventor)

Contents: 2 drive and belt tension, 3 shafts, bearings and diverter, 4 sensing and sort timing, 5 sort simulation, 6 frame and stability, 7 assembly checks, design iterations, assumptions.

2 Belt tension and drive

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Case	Te (N)	T shaft (Nm)	PWM	I (A)	SF stall	SF slip	SF coupling	T1 (N/mm)
normal (min pitch, 1 item diverting)	30.8	0.953	80 %	1.52	7.04	4.86	5.12	1.08
start-up (normal load, 0.5 s ramp)	31.4	0.974	80 %	1.55	6.89	4.76	5.01	1.08
accumulation jam (7 items held, belt slipping)	92.5	2.867	100 %	4.07	2.34	1.62	1.70	1.23

Te = (1 + 5 % flex) x (mu_slider (m_items + m_belt) g + slipping items mu_item m g + diverted-item drag) + bearing seal drag + m_eq a. Pretension 200 N per strand (0.2 % install elongation x 5 N/mm/% x 200 mm) carries Te up to 149 N without slip on a bare pulley (e^(mu pi) = 2.19). Take-up +/-15 mm covers 0.94 % belt elongation vs 0.2 % install. Full-voltage belt speed at normal load 0.326 m/s. Coupling rating 4.88 Nm is 0.73 x the 6.71 Nm motor stall: firmware must cut the motor on an encoder stall (see iterations).

3 Shafts, bearings and diverter

Check	Value	Margin
Radial load on the pulley (belt 2 T0 + weight)	400 N	-
Bending at the hub, 20 mm 1045 CD	19.7 MPa	M 15.5 Nm
Static von Mises at motor stall (20 / 15 mm)	21.1 MPa	SF 25.1 / 30.2
Fatigue, DE-Goodman (Kf 2.0, Se 230 MPa)	-	SF 5.75
Mid-span deflection / slope at the bearing	0.140 mm / 0.076 deg	UC insert allows 2 deg
UCFL204 L10 at 77 rpm	5.67e+07 h	static SF 33.2
Item sliding along the paddle (mu_f 0.25)	0.151 m/s	diverts below 76 deg
Paddle normal force, 1.5 kg item	7.14 N	belt drag 5.96 N
Servo hold torque (r 320 mm)	2.29 Nm	SF 2.15 vs 4.9 Nm
Servo impact torque (20 ms carton impact)	3.28 Nm	SF 1.49
Servo move 40 deg (speed + inertia limit)	0.109 s	design 0.15 s
Paddle 6061 3 mm: stress / impact / deflection	18.7 / 25.2 MPa	SF 10.9, 3.4 mm
Push-off time (CoM 103 mm sideways)	1.06 s	-

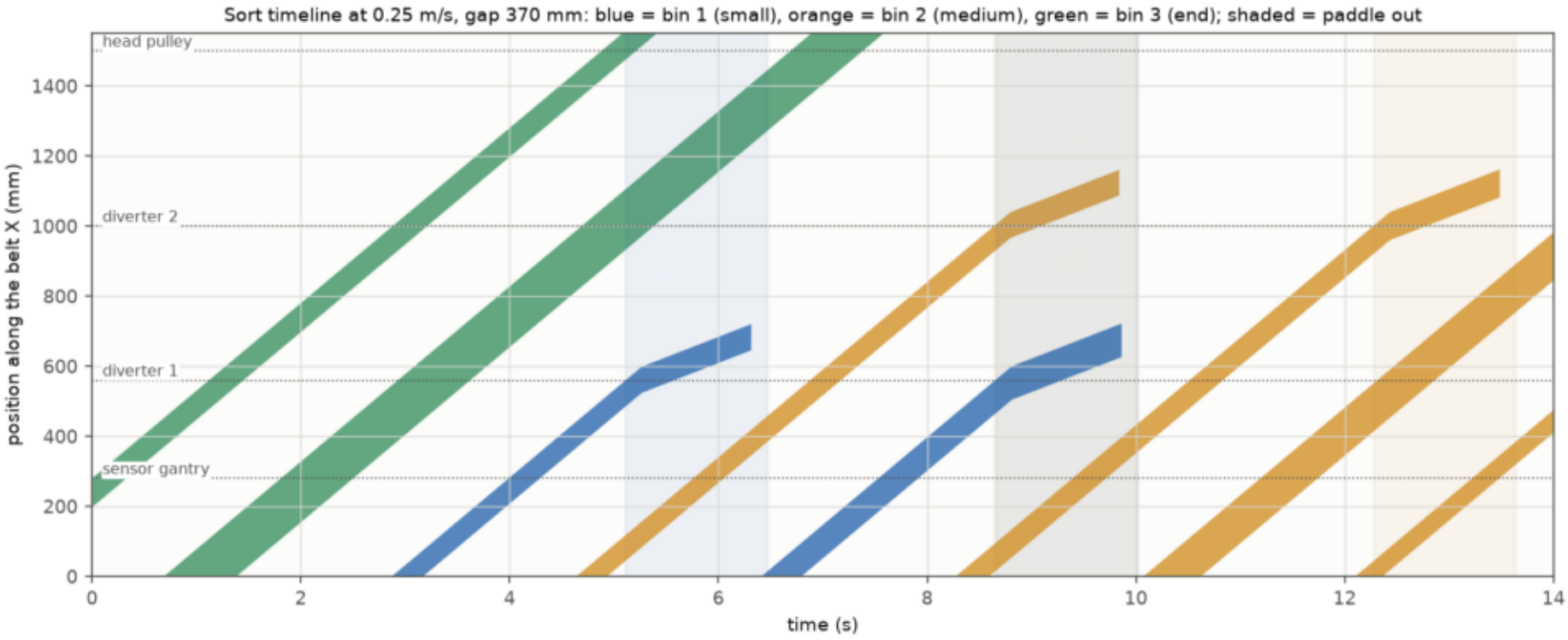
4 Sensing and sort timing

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Through-beam at 15 mm above the belt sees every item (H_min 30 mm). Length = belt encoder counts while the beam is blocked: 0.137 mm/count, 1 ms response -> sigma 0.28 mm. Height = 207.8 mm - ToF range (VL53L1X, 10+ samples per item at 50 Hz); range at the tallest item 57.8 mm > 40 mm minimum.

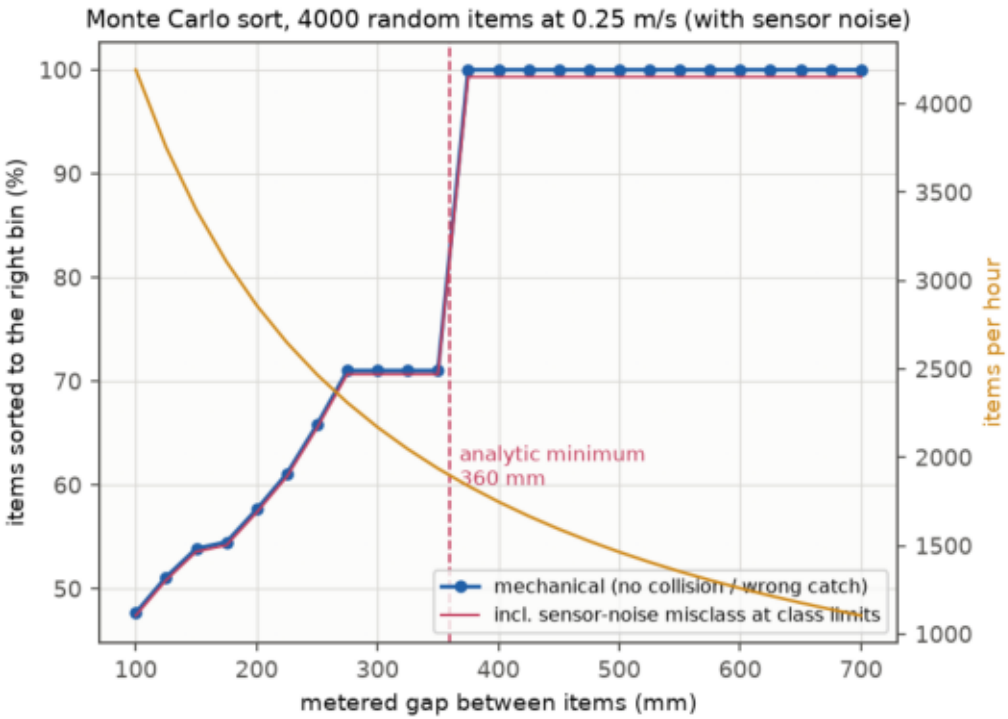
Decision: the item tail must clear the beam, the Arduino classifies (10 ms) and the paddle must be fully out (0.15 s) before the item reaches the first contact point X 582.6 mm (item hugging the pivot-side guide). With the 180 mm longest item the margin is 0.33 s.

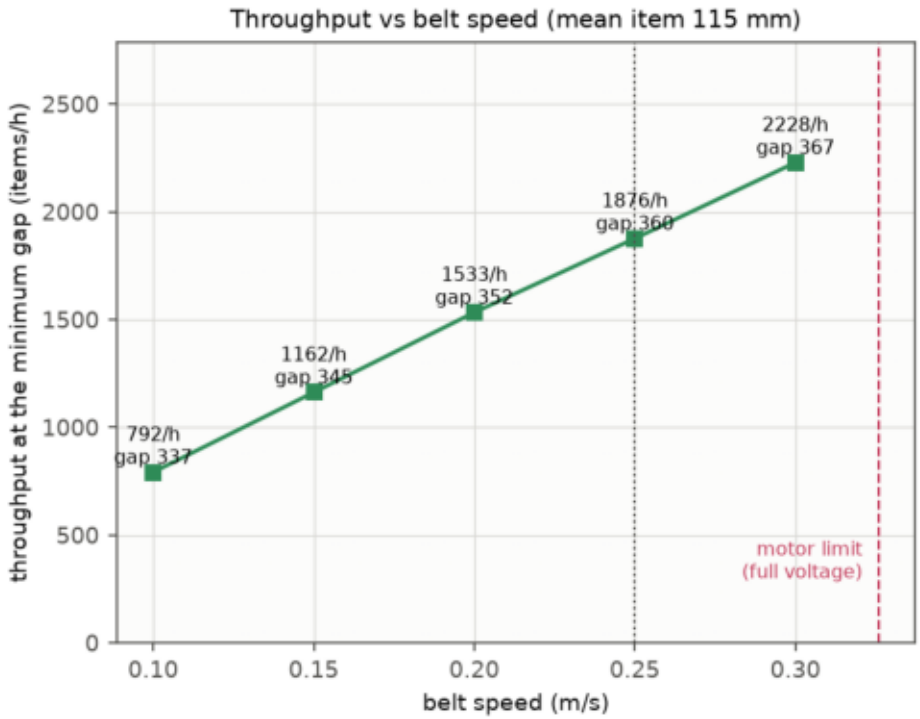
Minimum gap, item ahead passing straight on: its tail must be past the paddle sweep (X 905) when the paddle starts to move, so gap >= 905 - 582.6 + V t_deploy = 360 mm. Item behind a diversion: the push-off (1.06 s) plus stow needs 325 mm pitch = 275 mm gap for a 50 mm item. Design gap 370 mm (infeed metering, not modelled).



5 Sort simulation (Monte Carlo, 4000 random items)

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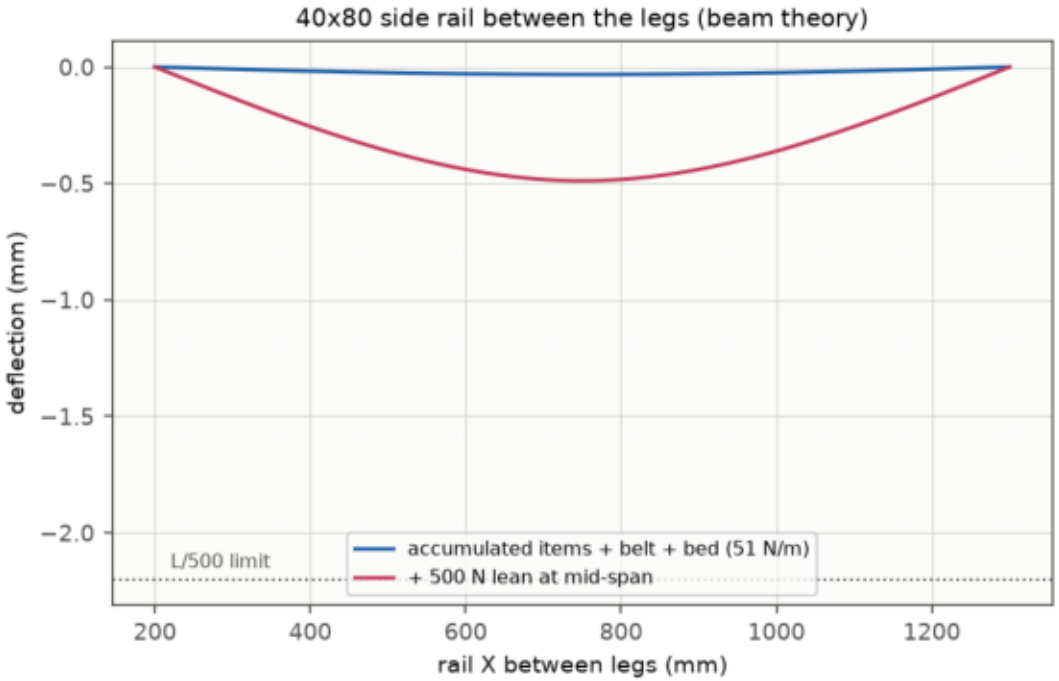


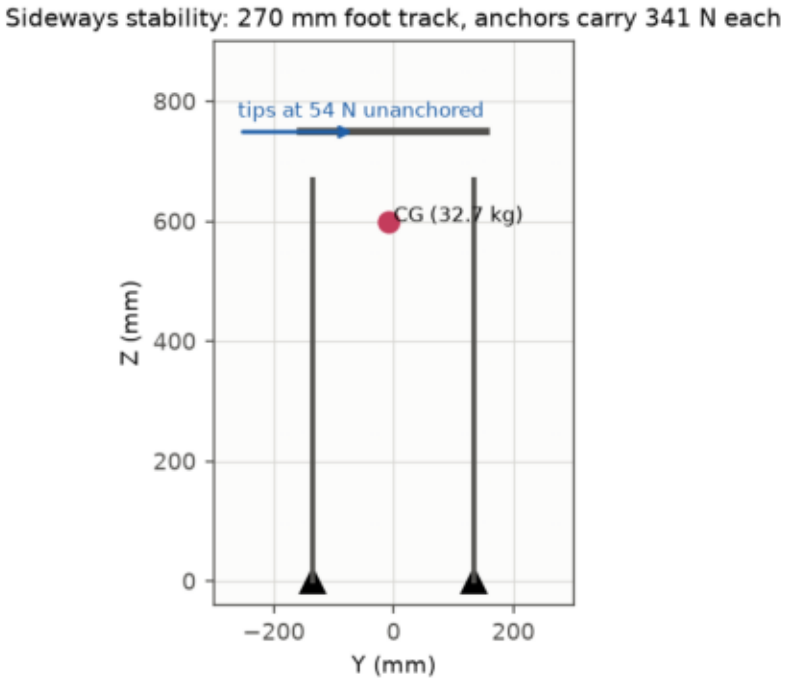
V (m/s)	min gap (mm)	items/h	right bin
0.10	337	792	99.48 %
0.15	345	1162	99.45 %
0.20	352	1533	99.38 %
0.25	360	1876	99.33 %
0.30	367	2228	99.30 %

At 0.25 m/s and a 370 mm gap: 1858 items/h, mechanical success 100.0 % (sweep collisions 0, wrong catches 0, late decisions 0). 27 of 4000 items (0.68 %) land in the neighbouring class because their size lies within ~1 mm of a class limit (sensor noise). The sweep finds 100 % mechanical success from 375 mm (25 mm grid) - the analytic minimum is 360 mm. Class shares (uniform sizes): small 9 %, medium 41 %, large 50 %. Throughput scales with speed; the motor limits the belt to 0.33 m/s at normal load.

6 Frame, stability, chutes and totes

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Check	Value	Note
Rail 40x80 deflection, jam load + 500 N lean	0.49 mm	limit L/500 = 2.2 mm
Sideways push that tips the unanchored frame	54 N	too low -> anchor the feet
Anchor uplift, 300 N side push at belt height	341 N per anchor	SF 11.7 on M8 wedge anchors
Chute 35 deg vs carton friction 0.40	a = 2.41 m/s2	exit 1.20 m/s
Tote fill time (bin 1 / 2 / 3)	56 / 5 / 3 min	170 / 71 / 41 items per tote

7 Assembly checks, design iterations, assumptions

Inventor checks: 104 occurrences (42 files), 309 constraints {'Flush': 249, 'Mate': 58, 'Angle': 2}, total DOF 0, 0 unhealthy. Interference: 0 pairs at the design pose, 5 diverter poses (20 / 40 deg each and both) and 4 take-up positions (-15 to +15 mm, pulley centres 1485-1515 mm). Paddle tips at 40 deg measured at [824.29, -99.76] / [1264.29, 99.76] mm = design. Overhead clearance over the item channel: lowest part ToFBoard at 207.84 mm (items up to 150 mm).

Design iterations driven by these checks:
1. Overhead clearance (iteration 1 -> 2): the ToF sensor face sat 127.8 mm above the belt, below the 150 mm item height limit (tall cartons would have hit it), and the gantry beam left 1 mm. The VL53L1X also needs 40 mm minimum range, so the gantry was raised 80 mm: 230 mm posts, ToF face 207.84 mm above the belt (57.84 mm range at the tallest item). Re-checked: 0 DOF, 0 clashes.
2. Frame stability: the 270 mm foot track tips sideways at 54 N at belt height -> the drawing requires 4 x M8 floor anchors (uplift 341 N each at a 300 N push, SF 12).
3. Coupling: the L-070 jaw coupling (4.88 Nm) is below the 6.71 Nm gearmotor stall. Jam torque is 2.87 Nm (SF 1.70), so it is kept with an encoder stall cut-off in firmware; an L-075 (10.2 Nm) is the drop-in upgrade if the stall case must be carried.
4. Part numbers: the gantry post shared CS-311 with the servo, which merged them in the parts list -> post renumbered CS-313.

Assumptions: belt class values (2.4 kg/m2, 5 N/mm at 1 %, mu 0.30 slider / 0.25 pulley / 0.50 carton), goBILDA 5203 datasheet (117 rpm, 6.71 Nm, 9.2 A) with an assumed 2e-6 kg m2 rotor and face pattern, Savox SB-2290SG 50 kg cm at 7.4 V, 0.15 s deploy incl. latency, 20 ms carton impact, ToF sigma 3 mm per sample, 40x80 rail Ix 44 cm4, UC seal drag 0.03 Nm, items uniformly distributed in size and centred between the guides. Not modelled: infeed metering, electrical wiring, controls firmware, belt splice, chute friction variation, FEA (stresses are low enough that beam theory governs: highest stress 25 MPa in the paddle).